

EXPERIMENT 1

Linux Programming Environment and Basic Commands

Name: Yug Nirwal

SAP ID: 59003354

Subject: C Programming

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Aim

To set up a Linux environment using Kali Linux in Oracle VirtualBox and practice some basic Linux commands used for programming.

1. Linux Programming Environment

We can write and run C programs using either an online Linux terminal or a Linux system installed on our computer.

Online method

For online practice, CoCalc can be used. It gives us a Linux terminal in the browser, so we can write, compile, and run C programs without installing Linux.

? For example: nano file.c, cc file.c and ./a.out

Offline methods

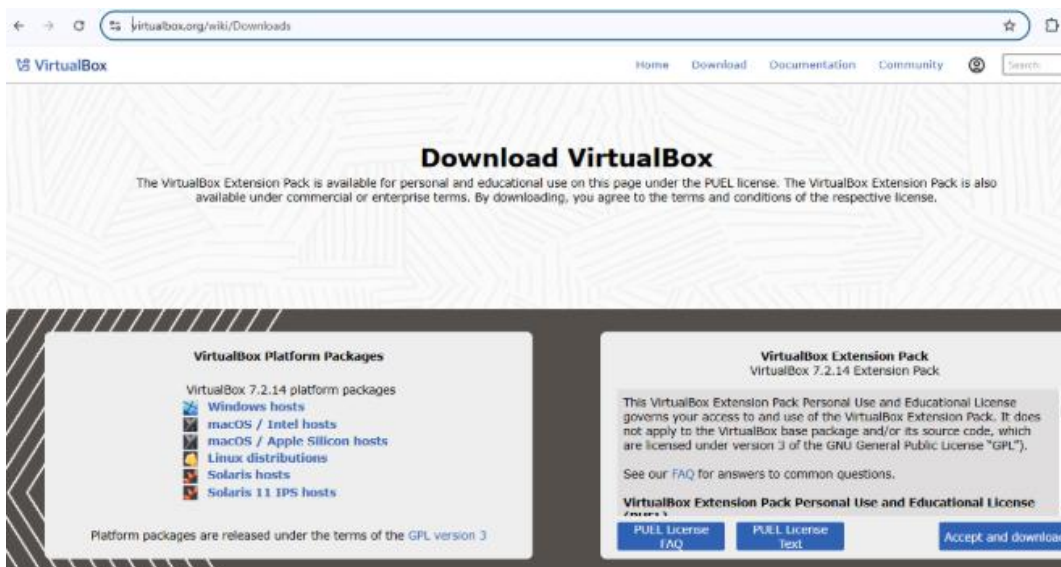
- ? Dual Boot - Windows and Linux are both installed, and we choose one when the computer starts.
- ? Single Boot - Only Linux is installed on the computer.
- ? Bootable USB - Linux can be started from a pen drive without changing the main operating system.
- ? Virtual Machine - Linux runs inside the current operating system using virtualization software.

Virtual Machine Manager (Hypervisor)

A hypervisor is software that helps us run a virtual machine. Type-1 runs directly on the hardware, while Type-2 runs inside an existing operating system. In this experiment, I used Oracle VirtualBox, which is a Type-2 hypervisor, to run Kali Linux.

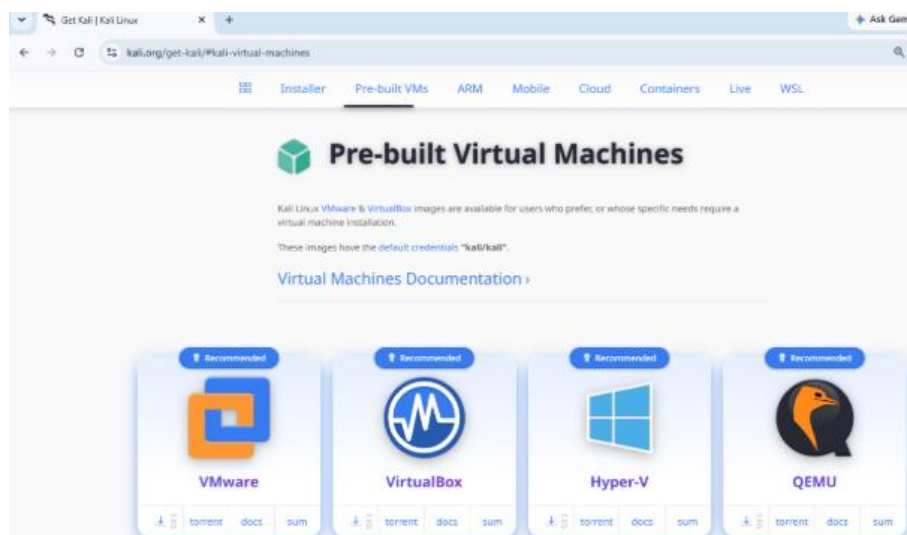
2. Installing Oracle VirtualBox and Kali Linux

Step 1: Download Oracle VirtualBox for the computer.



Oracle VirtualBox download page

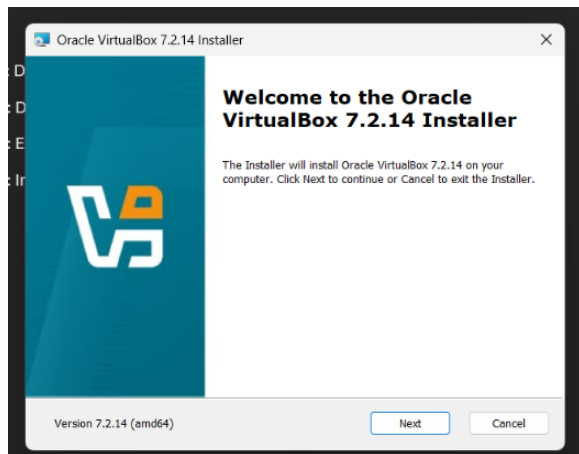
Step 2: Download the pre-built Kali Linux VM for VirtualBox.



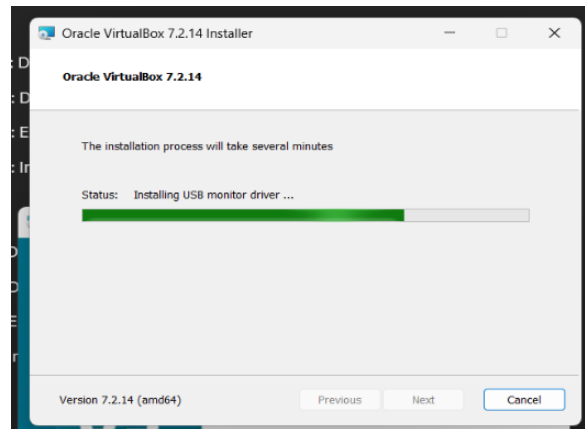
Kali Linux pre-built virtual machine download page

Step 3: Extract the downloaded Kali Linux files.

Step 4: Install Oracle VirtualBox by following the setup steps.

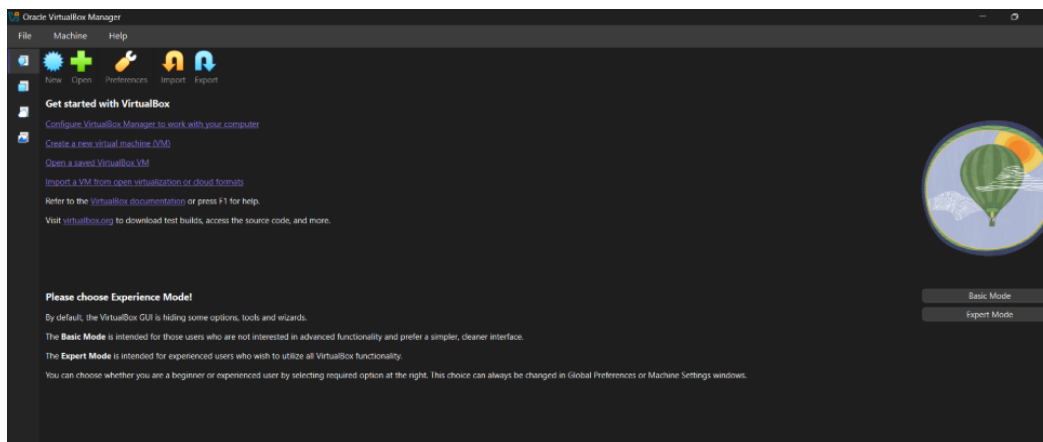


VirtualBox setup wizard



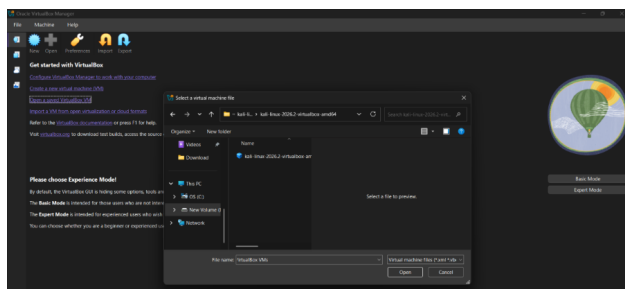
VirtualBox installation in progress

Step 5: Open VirtualBox after the installation is complete.

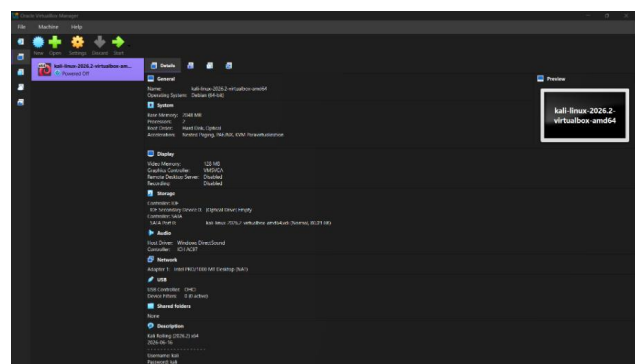


Oracle VirtualBox Manager

Step 6: Open the saved Kali Linux VM in VirtualBox.

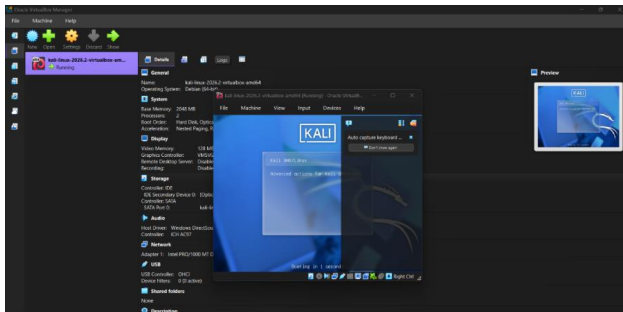


Selecting the saved VM

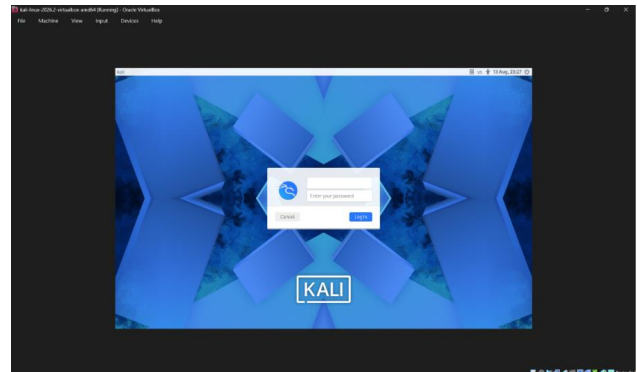


Kali Linux VM loaded in VirtualBox

Step 7: Start the VM and wait for Kali Linux to open.



Starting the Kali Linux VM



Kali Linux login screen

Step 8: Open the terminal in Kali Linux and start practicing commands and C programs.

3. Basic Linux Commands

I practiced the following commands to move around folders, create or remove directories, check previous commands and work with files.

Command	Purpose	Example
pwd	Shows the folder I am currently in.	pwd
mkdir	Creates a new folder.	mkdir testfolder
rmdir	Removes an empty folder.	rmdir testfolder
ls	Shows files and folders in the current location.	ls
cd	Moves into another folder.	cd Desktop
cd ..	Goes back one folder.	cd ..
clear	Clears the terminal window.	clear
history	Shows commands used earlier.	history
man	Shows help/manual for a command.	man ls
whoami	Shows the current user name.	whoami

File handling commands

Command	Purpose	Example
vi	Opens a file for creating or editing.	vi mynotes.txt
file	Shows the type of a file.	file file1.txt
rm	Removes a file.	rm file1.txt
cp	Makes a copy of a file.	cp file1.txt file2.txt

mv	Moves a file or changes its name.	mv file2.txt renamed.txt
cat	Shows all content of a file.	cat file1.txt
more	Shows file content one page at a time.	more file1.txt
less	Lets us scroll through file content.	less file1.txt
head	Shows the first few lines of a file.	head file1.txt

To save a file in vi, press Esc, type :wq and then press Enter. This saves the file and closes the editor.

Result

Oracle VirtualBox and Kali Linux were set up successfully. I also practiced basic Linux directory and file commands in the terminal. After this experiment, the Linux environment was ready to use for writing, compiling and running C programs.